Sydney Retail Customer Behaviour Analysis

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Import the data for all of the questions

```r
retail = read.csv("Retail.csv")
```

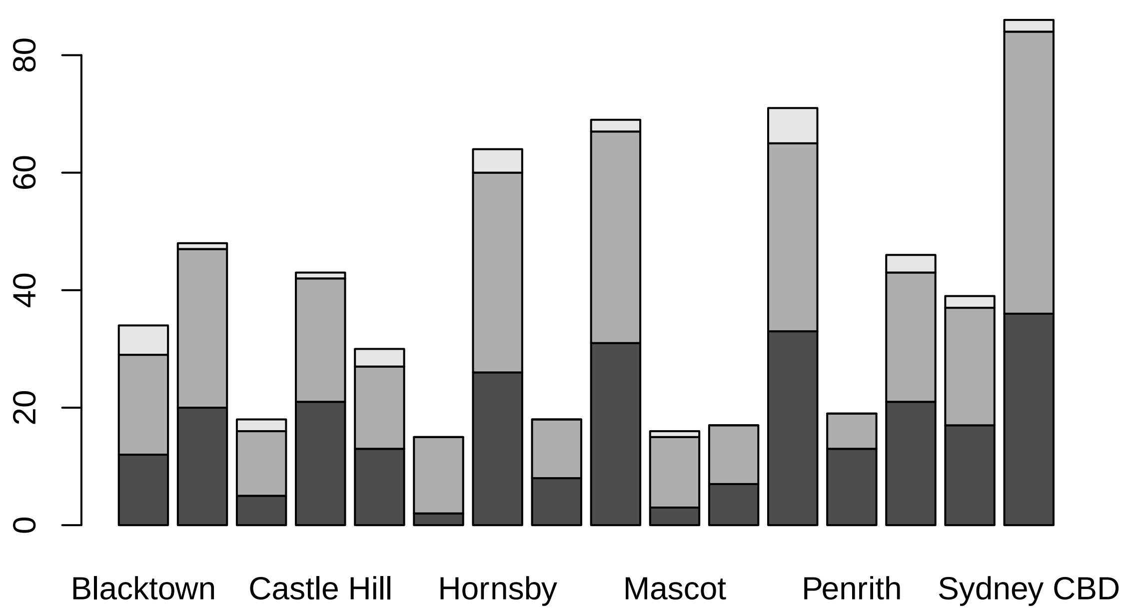
A. Gender and Store Location Independence

a) Test whether store location is independent of gender.

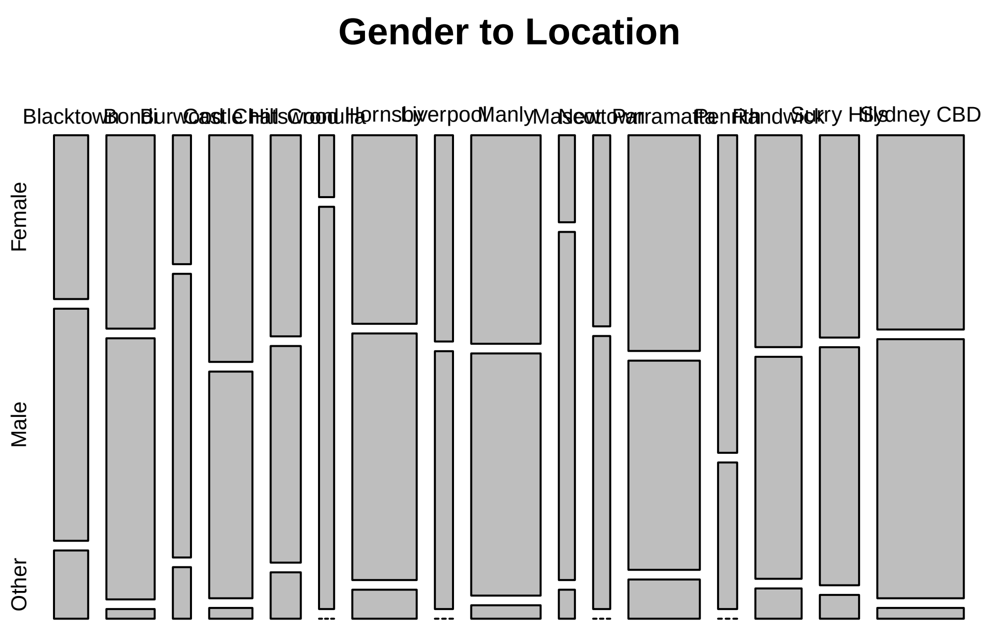
1. Visualize the data

I use stacked bar plot to visualize categorical to categorical data

```r
barplot(table(retail$Gender,retail$StoreLocation))
```



```r
plot(table(retail$StoreLocation, retail$Gender), main = "Gender to Location" )
```




```r
table(retail$Gender, retail$StoreLocation)
```

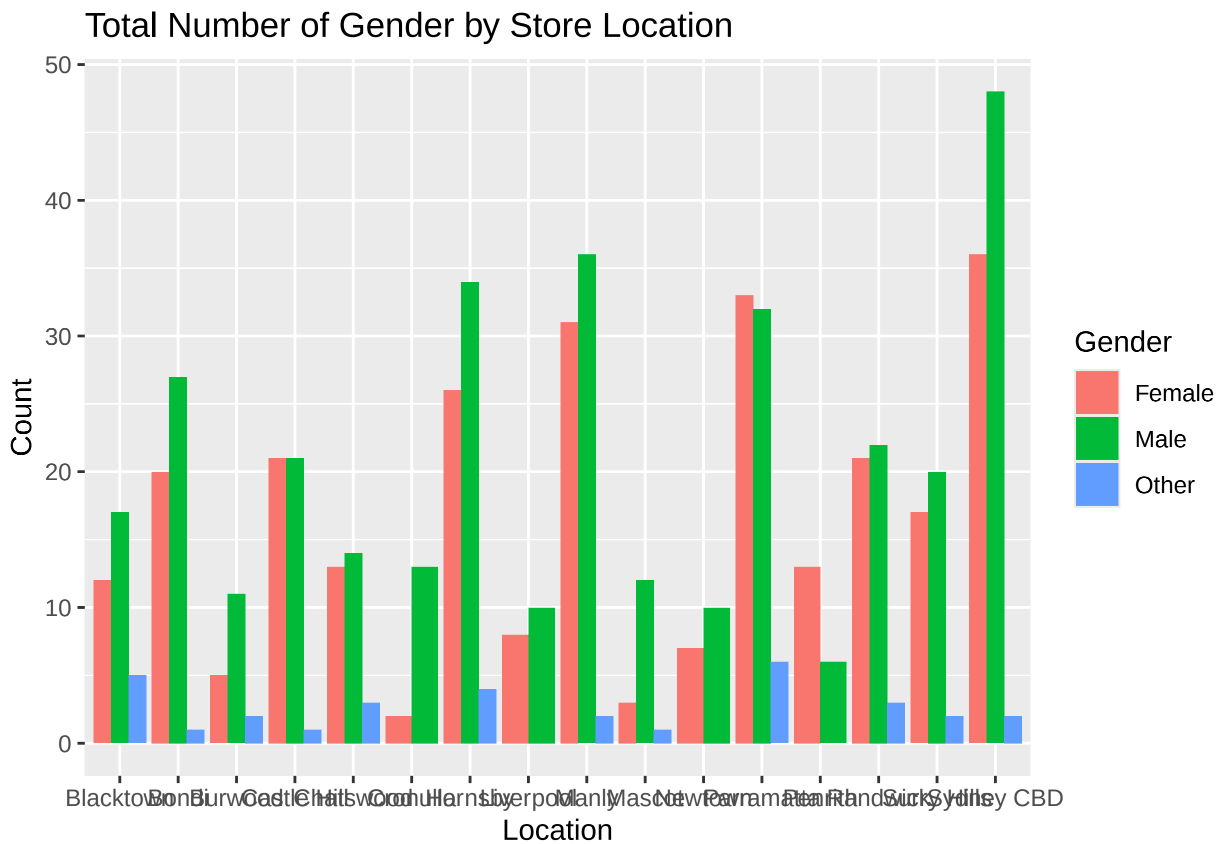
```
##
##          Blacktown Bondi Burwood Castle Hill Chatswood Cronulla Hornsby
##   Female        12    20       5          21        13        2      26
##   Male          17    27      11          21        14       13      34
##   Other          5     1       2           1         3        0       4
##
##          Liverpool Manly Mascot Newtown Parramatta Penrith Randwick Surry Hills
##   Female         8    31      3       7         33      13       21          17
##   Male          10    36     12      10         32       6       22          20
##   Other          0     2      1       0          6       0        3           2
##
##          Sydney CBD
##   Female         36
##   Male           48
##   Other           2
```

Or I can use ggplot to plot this data.

```r
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```r
ggplot(retail, aes(x = StoreLocation, fill = Gender)) + #data in the plot from retail, x axis storeloca
  geom_bar(position = "dodge") + #Position = "dodge" is to split the bar chart into each gender for eac
  labs(title = "Total Number of Gender by Store Location", x = "Location", y = "Count") #Name the axis
```



From the bar plot it seems that overall store has more male customers compare to female and other, except in Penrith.

2. Show all the key steps of hypothesis testing

Step 1: Explore the data

```r
table(retail$Gender)
```

```
##
## Female   Male  Other
##    268    333     32
```

```r
table(retail$StoreLocation)
```

```
##
##    Blacktown        Bondi      Burwood Castle Hill    Chatswood     Cronulla
##           34           48           18          43           30           15
##      Hornsby    Liverpool        Manly      Mascot      Newtown   Parramatta
##           64           18           69          16           17           71
```

```
##     Penrith    Randwick Surry Hills  Sydney CBD
##          19          46          39          86
```

Step 2: Define hypothesis

H0: Store location is independent of gender

Ha: Store location is not independent of gender

Step 3: Do chi-square test and look for p-value

With build in chisq.test:

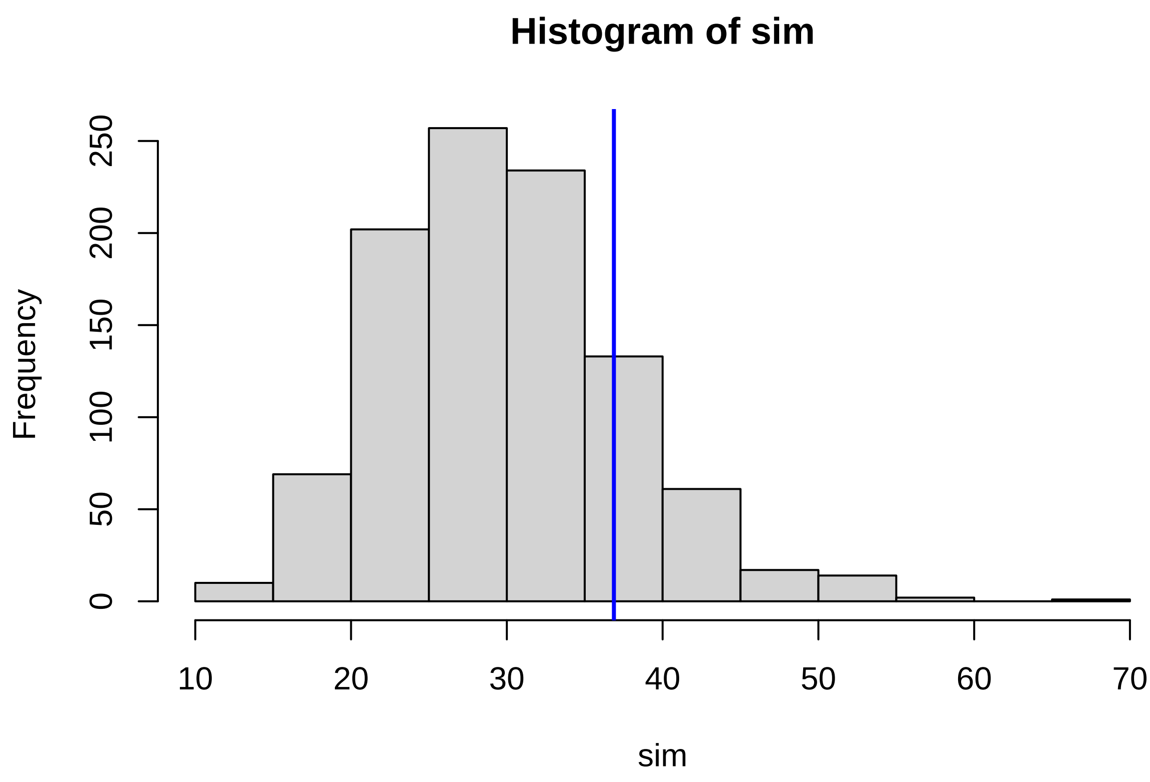
```
chisq_result = chisq.test(table(retail$StoreLocation, retail$Gender), simulate.p.value = TRUE, B=2000)
chisq_result$expected
```

```
##
##                  Female      Male     Other
##   Blacktown   14.394945 17.886256 1.7187994
##   Bondi       20.322275 25.251185 2.4265403
##   Burwood      7.620853  9.469194 0.9099526
##   Castle Hill 18.205371 22.620853 2.1737757
##   Chatswood   12.701422 15.781991 1.5165877
##   Cronulla     6.350711  7.890995 0.7582938
##   Hornsby     27.096367 33.668246 3.2353870
##   Liverpool    7.620853  9.469194 0.9099526
##   Manly       29.213270 36.298578 3.4881517
##   Mascot       6.774092  8.417062 0.8088468
##   Newtown      7.197472  8.943128 0.8593997
##   Parramatta  30.060032 37.350711 3.5892575
##   Penrith      8.044234  9.995261 0.9605055
##   Randwick    19.475513 24.199052 2.3254344
##   Surry Hills 16.511848 20.516588 1.9715640
##   Sydney CBD  36.410742 45.241706 4.3475513
```

Without build in chisq.test (I make it into 1 chunk of code to prevent excessive number of sections)

```
#Create a table to store Gender and Store Location only

a = table(retail$Gender,retail$StoreLocation)

#Calculate the expected value
n=sum(a) #Total customers

gender.sum = rowSums(a) #Female, male, Other
storeloc.sum = colSums(a) #Total customer on each store
e = outer(gender.sum, storeloc.sum)/n #Expected value

#Difference between real data and expected value

retail.chisq = sum((a-e)^2/e) #Use this value to calculate p-value after simulation

#Simulation
location = colnames(a) #get the name of the columns from table a
gender = rownames(a) #get the name of rows from table a
```

```r
sim = replicate(1000,{
  #sample the set of location
  samp.loc = sample(location, size = n, replace = TRUE, prob = storeloc.sum/n)
  #sample the set of gender
  gen.loc = sample(gender, size = n, replace = TRUE, prob = gender.sum/n)
  x = table(gen.loc, samp.loc)

  #compute expected value
  gen.sim = rowSums(x)
  store.sim = colSums(x)
  expected.sim = outer(gen.sim,store.sim)/sum(x)

  #compute the difference
  sum((x-expected.sim)^2/expected.sim)
})

hist(sim)
abline(v = retail.chisq, lwd = 2, col = "blue")
```




```r
#Calculate the p-value
sum(sim>retail.chisq)/1000 #divided by 1000 because I did 1000 simulation
```

```
## [1] 0.175
```

Resultant P-value is similar between in build and simulation test. P-value is greater than significant value (0.05).

Step 4: Conclusion based on the p-value obtained

P-value larger than 0.05, so I do not have enough evidence to reject null hypothesis. So, store location is independent of gender.

3. Interpretation

After I did chi-square test, it showed that store is independent of gender or I can say that gender does not affect store location. It means that male, female, and other does not have different preferences for shopping locations. So, my suggestions is to make the stores appeal to all of the genders not only to certain gender, by the interior design and marketing campaign that provide in-store promotions not only to certain gender product.

B. Purchase Amount Comparison: Manly vs Parramatta

Compute a 95% confidence interval for the difference in the mean purchase amount between customers who shop in Parramatta and those who shop in Manly?

1. Visualize the data

```
#Obtain the exact data that we need to observe (Data in Parramatta and Manly)
manly = subset(retail, subset = (retail$StoreLocation == 'Manly')) #take data located in Manly only
parramatta = subset(retail, subset = (retail$StoreLocation == 'Parramatta')) #take data located in Parr

#combined data
combined_data = retail[retail$StoreLocation %in% c("Parramatta", "Manly"), ]

#To visualize numerical to categorical data I can use boxplot

boxplot(PurchaseAmount ~ StoreLocation, data = combined_data, col = c("cyan", "pink"),
        main = "Difference in Purchase Amounts between Parramatta and Manly",
        xlab = "Store Location", ylab = "Purchase Amount")
```

From the plot we can see Manly purchase amounts have higher variances compare to Parramatta. Median value of purchases amounts are similar in both shops. From the box plot seems Manly has high purchase amount outlier, while Parramatta has low purchase amounts outlier.

2. Do T-test to compare different means in two groups

Step 1. Define the hypothesis

H0: There are no difference in means of purchase amount between Manly and Parramatta

Ha: There are difference in means of purchase amount between Manly and Parramatta

Step 2. Explore the data

```r
str(manly)
```

```
## 'data.frame':    69 obs. of  16 variables:
##  $ customerID          : int  124657 124666 124668 124678 124687 124697 124731 124737 124748 124757
##  $ Age                 : int  64 66 61 72 70 65 36 61 73 72 ...
##  $ Gender              : chr  "Male" "Female" "Female" "Male" ...
##  $ AnnualIncome        : int  90 100 83 201 238 73 65 56 237 231 ...
##  $ PurchaseAmount      : int  110 69 133 211 233 95 47 148 253 226 ...
##  $ StoreLocation       : chr  "Manly" "Manly" "Manly" "Manly" ...
##  $ NumberOfItems       : int  31 16 26 63 55 17 27 26 73 58 ...
##  $ PaymentMethod       : chr  "Mobile Payment" "Mobile Payment" "Debit Card" "Credit Card" ...
##  $ SatisfactionScore   : int  6 8 7 8 7 6 5 5 8 6 ...
##  $ LoyaltyProgram      : chr  "No" "Yes" "Yes" "Yes" ...
##  $ ShoppingFrequency   : chr  "Weekly" "Weekly" "Daily" "Weekly" ...
```

```
## $ GroceryPurchase     : int  0 0 0 1 1 1 1 0 1 1 ...
## $ HouseholdPurchase   : int  1 0 0 1 0 1 1 1 1 0 ...
## $ PersonalCarePurchase: int  0 1 1 1 1 1 1 1 1 1 ...
## $ BabyPurchase        : int  1 1 0 0 0 1 0 1 0 0 ...
## $ PetPurchase         : int  0 0 0 1 0 0 0 1 1 1 ...
```

```r
str(parramatta)
```

```
## 'data.frame':    71 obs. of  16 variables:
## $ customerID          : int  124662 124664 124667 124673 124688 124694 124706 124708 124710 124725
## $ Age                 : int  37 45 22 23 26 34 48 49 54 53 ...
## $ Gender              : chr  "Female" "Female" "Female" "Female" ...
## $ AnnualIncome        : int  98 134 122 90 118 73 142 121 55 79 ...
## $ PurchaseAmount      : int  98 159 40 167 148 125 145 159 182 138 ...
## $ StoreLocation       : chr  "Parramatta" "Parramatta" "Parramatta" "Parramatta" ...
## $ NumberOfItems       : int  11 27 24 23 34 24 1 17 14 15 ...
## $ PaymentMethod       : chr  "Gift Card" "Credit Card" "Gift Card" "Mobile Payment" ...
## $ SatisfactionScore   : int  6 5 9 5 6 7 6 7 6 6 ...
## $ LoyaltyProgram      : chr  "Yes" "Yes" "No" "Yes" ...
## $ ShoppingFrequency   : chr  "Daily" "Weekly" "Weekly" "Weekly" ...
## $ GroceryPurchase     : int  1 1 1 0 1 0 0 0 0 0 ...
## $ HouseholdPurchase   : int  0 0 0 0 0 0 1 0 0 1 ...
## $ PersonalCarePurchase: int  0 1 1 1 0 1 0 1 0 0 ...
## $ BabyPurchase        : int  1 0 1 1 1 1 0 0 1 1 ...
## $ PetPurchase         : int  0 0 0 1 0 0 1 0 0 0 ...
```

Manly has fewer number of transactions. Sample is large enough, so I can do the test with in build function. Two sided test (just want to see if there is any difference or not)

Step 3. Do the test

```r
ttest = t.test(manly$PurchaseAmount, parramatta$PurchaseAmount, alternative = "two.sided")
ttest
```

```
##
##  Welch Two Sample t-test
##
## data:  manly$PurchaseAmount and parramatta$PurchaseAmount
## t = 3.3788, df = 104.97, p-value = 0.001023
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  14.20028 54.53864
## sample estimates:
## mean of x mean of y
##  149.1159  114.7465
```

Analyze the result pvalue = 0.001023 95 Confidence interval: 14.20028 54.53864 (Answer for the question)

- Interpret the result in no more than 3 sentences.

Step 4. Conclusion Based on the p-value I got from the test, I have enough evidence to reject null hypothesis. It means, there are difference in mean of purchase amounts between customers who shop in

Parramatta and customer who shop in Manly. So, I suggest to place more premium products in Manly (fewer number of transactions but higher mean of purchase amount compared to Parramatta), because the customer purchasing power seems higher in Manly rather than Parramatta.

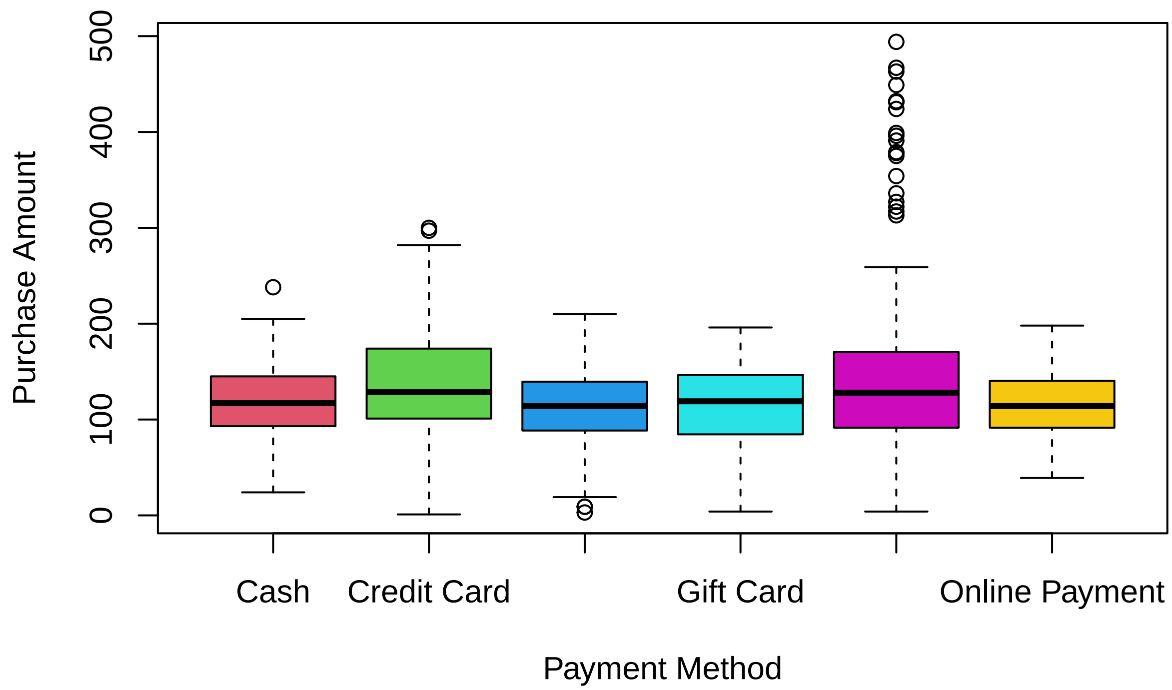
C. Payment Method and Purchase Amount

c) Test whether the purchase amount varies significantly across different payment methods.

Identify which specific payment methods show significant differences and explain how these insights could be used to make informed business decisions (only print the significant results).

1. Visualize the data

```
boxplot(retail$PurchaseAmount~retail$PaymentMethod, xlab = "Payment Method", ylab = "Purchase Amount", c
```



2. Show all the key steps of hypothesis testing

Step 1: Define hypothesis

H0: Mean purchase amount is the same across all payment methods.

H1: Mean purchase amount is not the same across all payment methods.

Step 2: Do permutation

```r
x = replicate(1000, {
  #Shuffle the payment methods to force the populations means to be equal
  method.perm = sample(retail$PaymentMethod)
  #Compute the F-statistic
  oneway.test(PurchaseAmount ~ method.perm, data = retail, var.equal = TRUE)$statistic
})
```

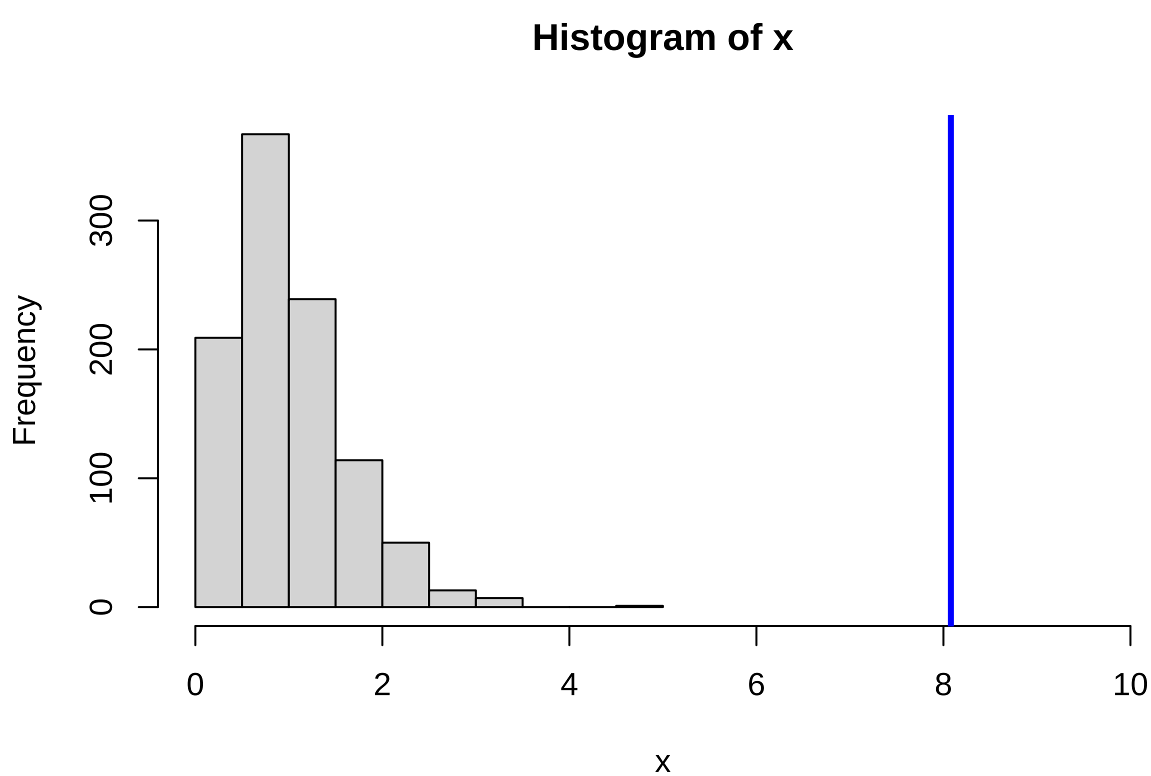
Step 3: Calculate F-statistic (actual)

```r
Fstat = oneway.test(PurchaseAmount~PaymentMethod, data = retail, var.equal = TRUE)$statistic
Fstat
```

```
##        F
## 8.079431
```

Step 4: Calculate p-value and create the histogram distribution of x

```r
hist(x, xlim = c(0,10))
abline(v=Fstat, col = "blue", lwd = 3)
```




```r
#Calculate the p-value
#Because F-statistic range between 0 to inf (positive), it is only one tail

pval = mean(x>Fstat)
pval
```

```
## [1] 0
```

- Interpret the result (Step 5. Conclusion)

The p-value is zero which is below 0.05. That is why I have evidence to reject null hypothesis. So, at least there is one pair of means purchase amount that are not the equal across all payment methods.

4. identify which specific payment methods show significant differences and explain how these insightscould be used to make informed business decisions.

After we know that the p-value is small, we can do the post hoc pairwise comparison test to find which pair has different mean of purchase amount.

```r
str(retail) #sample is large
```

```
## 'data.frame':    633 obs. of  16 variables:
##  $ customerID          : int  124655 124656 124657 124658 124659 124660 124661 124662 124663 124664
##  $ Age                 : int  26 24 64 64 56 42 25 37 47 45 ...
##  $ Gender              : chr  "Female" "Male" "Male" "Female" ...
##  $ AnnualIncome        : int  45 86 90 70 76 117 79 98 17 134 ...
##  $ PurchaseAmount      : int  164 71 110 160 108 143 88 98 156 159 ...
##  $ StoreLocation       : chr  "Surry Hills" "Randwick" "Manly" "Bondi" ...
##  $ NumberOfItems       : int  24 28 31 19 15 32 32 11 30 27 ...
##  $ PaymentMethod       : chr  "Online Payment" "Debit Card" "Mobile Payment" "Online Payment" ...
##  $ SatisfactionScore   : int  6 6 6 5 6 7 6 6 6 5 ...
##  $ LoyaltyProgram      : chr  "Yes" "No" "No" "Yes" ...
##  $ ShoppingFrequency   : chr  "Monthly" "Daily" "Weekly" "Daily" ...
##  $ GroceryPurchase     : int  1 1 0 0 1 1 1 1 0 1 ...
##  $ HouseholdPurchase   : int  1 0 1 0 0 1 1 0 1 0 ...
##  $ PersonalCarePurchase: int  0 1 0 1 0 0 1 0 0 1 ...
##  $ BabyPurchase        : int  1 0 1 0 1 0 1 1 0 0 ...
##  $ PetPurchase         : int  0 0 0 0 1 1 0 0 1 0 ...
```

```r
table(retail$PaymentMethod) #sample is large
```

```
## 
##           Cash    Credit Card     Debit Card      Gift Card Mobile Payment 
##             53            190            123             67            140 
## Online Payment 
##             60
```

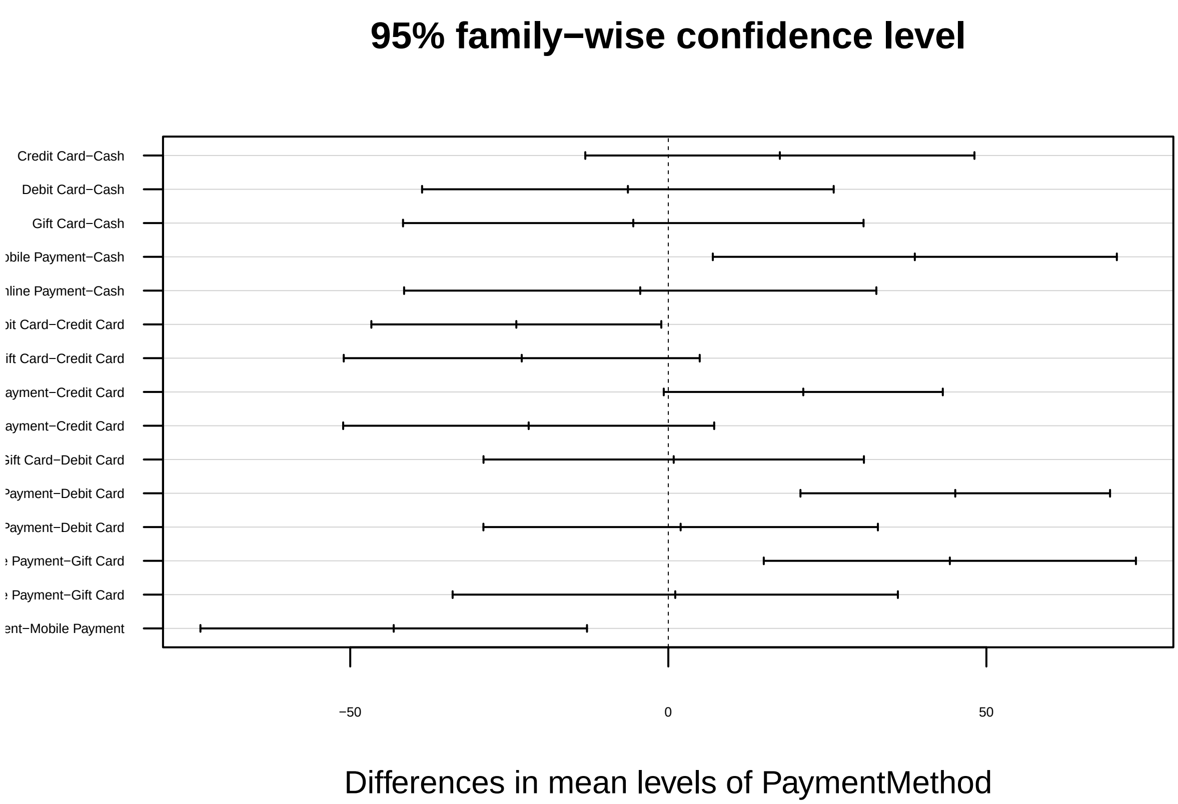
Sample is large enough. Now, I will do the post hoc test.

```r
fit = aov(PurchaseAmount~PaymentMethod, data = retail)

TukeyHSD(fit)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
## 
## Fit: aov(formula = PurchaseAmount ~ PaymentMethod, data = retail)
## 
```

```
## $PaymentMethod
##                                    diff         lwr        upr     p adj
## Credit Card-Cash              17.5339623 -13.0517331  48.119658 0.5730833
## Debit Card-Cash               -6.3546556 -38.7062584  25.996947 0.9933976
## Gift Card-Cash                -5.5063362 -41.7010582  30.688386 0.9980279
## Mobile Payment-Cash           38.7482480   6.9936470  70.502849 0.0068701
## Online Payment-Cash           -4.4160377 -41.5315637  32.699488 0.9993990
## Debit Card-Credit Card       -23.8886179 -46.6748633  -1.102372 0.0336131
## Gift Card-Credit Card        -23.0402985 -51.0160858   4.935489 0.1743016
## Mobile Payment-Credit Card    21.2142857  -0.7160919  43.144663 0.0645280
## Online Payment-Credit Card   -21.9500000 -51.1073155   7.207316 0.2621632
## Gift Card-Debit Card           0.8483194 -29.0479364  30.744575 0.9999995
## Mobile Payment-Debit Card     45.1029036  20.7701500  69.435657 0.0000024
## Online Payment-Debit Card      1.9386179 -29.0660670  32.943303 0.9999748
## Mobile Payment-Gift Card      44.2545842  15.0054031  73.503765 0.0002557
## Online Payment-Gift Card       1.0902985 -33.9057324  36.086329 0.9999992
## Online Payment-Mobile Payment -43.1642857 -73.5455129 -12.783059 0.0007778
```

```r
plot(TukeyHSD(fit),las=1, cex.axis=0.4)
```



From the plot it seems that Mobile Payment-Cash, Mobile Payment-Debit Card, Mobile Payment-Gift Card and Online Payment-Mobile Payment pairs have small p-value, it means that there is difference between those pairs in mean of purchase amounts.

- **Business Decisions**

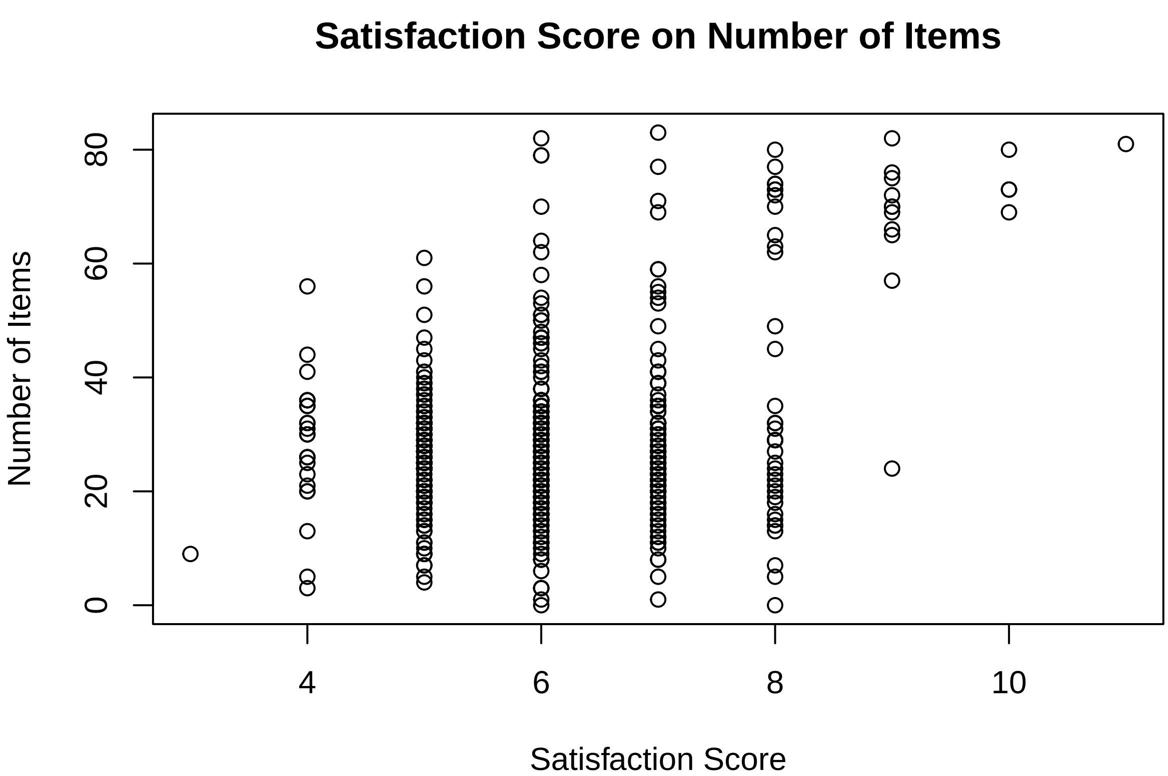
By comparing groups we can see several payment methods which customers like such as mobile payment rather than cash, mobile payment rather than debit card, and mobile payment rather than online payment. So, after get this insight I suggest to implement a program to increase sales / revenue for those who use mobile payments by giving discount for those who pay with mobile payments because most of transactions done by mobile payment.

D. Predictive Modelling: Satisfaction Score

d) Test if satisfaction score can be used to predict number of items. Assess the strength of the predictive power of satisfaction score for number of items.

1. Visualize the data

```
plot(retail$NumberOfItems~retail$SatisfactionScore, main = "Satisfaction Score on Number of Items",
     xlab = "Satisfaction Score", ylab = "Number of Items")
```



2. Do hypothesis testing

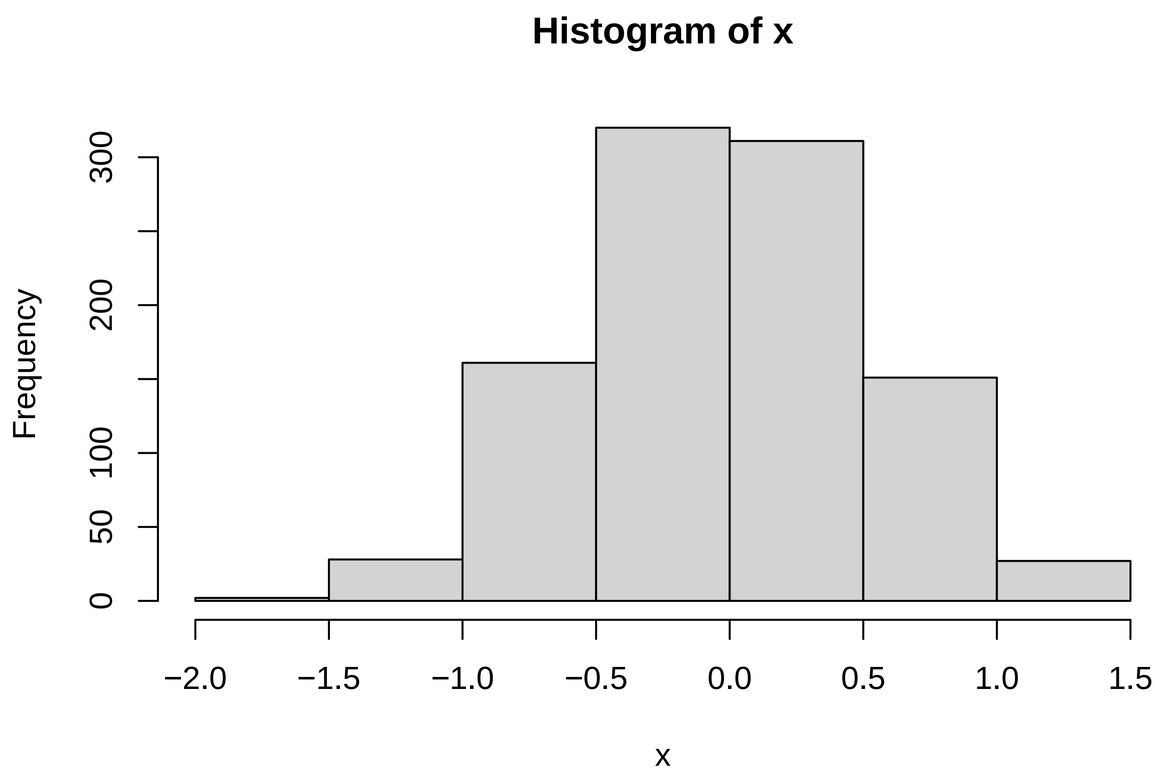
Step 1: State the hypothesis

H0: b = 0 (Slope is zero, meaning satisfaction score can not be used to predict number of items)

Ha: b != 0 (Slope is not zero, meaning satisfaction score can be used to predict number of items)

Step 2: Bootstrap

```r
x= replicate(1000, {
  score.perm = sample(retail$SatisfactionScore) # shuffle score to force population b = 0
  fit = lm(retail$NumberOfItems ~ score.perm, data=retail) # fit the straight line model
  coef(fit)[2] # return the slope value
})

#create histogram of x
hist(x)
```



Step 3: Calculate the b (slope) from data

```r
#Create the model
regression_model = lm(NumberOfItems~SatisfactionScore, data = retail)
slope = coef(regression_model)[2]
regression_model
```

```
##
## Call:
## lm(formula = NumberOfItems ~ SatisfactionScore, data = retail)
##
## Coefficients:
##       (Intercept)  SatisfactionScore
##             1.412              4.458
```

```r
summary(regression_model)
```

```
##
## Call:
## lm(formula = NumberOfItems ~ SatisfactionScore, data = retail)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -37.079  -8.620  -1.620   5.838  53.838
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)         1.4116     3.1941   0.442    0.659
## SatisfactionScore   4.4584     0.5163   8.635   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.74 on 631 degrees of freedom
## Multiple R-squared:  0.1057, Adjusted R-squared:  0.1043
## F-statistic: 74.55 on 1 and 631 DF,  p-value: < 2.2e-16
```

Number of Items = 1.412 + 4.458*Satisfaction Score

Step 4: Calculate the p-value

```r
p.value = mean(x > slope) + mean(x < (-slope))
p.value
```

```
## [1] 0
```

3. Interpret the results, both statistically and in the context of the case. (Conclusion)

Step 5: Conclusion

Because the p-value (0) less than significant value (0.05), so I reject the null hypothesis, which mean that Slope is not zero, means satisfaction score can be used to predict number of items.

4. predict number of items for a customer who scored the store 8

```r
pred_result = predict(regression_model, newdata = data.frame(SatisfactionScore = 8))
pred_result
```

```
##        1
## 37.07869
```

If the store got score 8 as the satisfaction score, then the store will have 37.1 (rounded) items. Based on the summary of the models, the value of R-squared of the model is 10.57% which mean only around 10 percent variation of number of items is explained by satisfaction score. This indicates that the model can be improved by adding other significant variables which affect total number of items in one store. **So, for conclusion this prediction due to low R-squared value is not dependable.**

5. Examine the residuals

```r
summary(regression_model)
```

```
##
## Call:
## lm(formula = NumberOfItems ~ SatisfactionScore, data = retail)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -37.079  -8.620  -1.620   5.838  53.838
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)         1.4116     3.1941   0.442    0.659
## SatisfactionScore   4.4584     0.5163   8.635   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.74 on 631 degrees of freedom
## Multiple R-squared:  0.1057, Adjusted R-squared:  0.1043
## F-statistic: 74.55 on 1 and 631 DF,  p-value: < 2.2e-16
```

residuals meaning how difference the actual value to the predicted value. The larger it is means that the model has a bad performance. The summary also provide the number of residual range -37.079 and 53.838, it means that sometime the model might under predict the number of items and over predict. So based on the summary, residual is large and the adjusted r-squared is very low that indicate only 10.43% variation of number of items can be explained by satisfaction score.

D. Sampling Methodology

Describe the potential sampling process that Aussie Prime might have used to collect this data. Consider how they might have ensured the sample is representative of their entire customer base and discuss any sampling methods that could have been employed.

Random Sampling

Aussie Prime can just pick one random data from the entire database. This method ensure that every customer in the database have an equal chance of being selected as the representative. This method also reduces bias that might happened.